REGULAR EXTERIOR COHERENCE AND INFINITESIMAL RIGIDITY OF WEAKLY CONVEX DECOMPOSABLE POLYHEDRA

BENJAMIN HUINDA

ABSTRACT. We introduce an exterior-coherence criterion for infinitesimal rigidity of weakly convex polyhedra. Given a weakly convex decomposable polyhedron $P \subset \mathbb{R}^3$ with convex hull $Q = \mathrm{conv}(V(P))$, we associate to the exterior closure $\mathcal{E}(P) = Q - P$ an active signed affine-circuit matrix $A_{\mathrm{ext}}(P)$. The regular exterior coherence condition is the strict feasibility condition

$$\exists h \qquad A_{\mathrm{ext}}(P)h > 0,$$

or, equivalently by Gordan's theorem, the absence of a non-zero positive dependence $y \geq 0$, $A_{\mathrm{ext}}(P)^T y = 0$. The main theorem proves that a weakly convex decomposable polyhedron satisfying regular exterior coherence is infinitesimally rigid. The proof replaces codecomposability by a positive exterior Schur correction. The local sign of this correction follows from the Hilbert–Einstein Hessian theorem of Izmestiev–Schlenker applied to a triangular bipyramid. The global coherence mechanism is expressed through stellar subdivision, conformal circuit elimination, positive height descent, and the positive tropical Spin_{12} Pfaffian chamber. We also isolate the remaining PL-topological bridge needed to recover the full Izmestiev–Schlenker conjecture from the criterion: the exterior closure of every weakly convex decomposable polyhedron must admit a derived regular relative PL model. A companion Python verification script verifies the finite rank-two packet exclusion and the formal dependency chain.

CONTENTS

1. Introduction 1
2. Weakly convex polyhedra and Hilbert–Einstein Hessians 2
3. Affine circuits and active exterior rows 3
4. The primitive rank-two packet 4
5. Stellar subdivision and positive height descent 4
6. Derived exterior regularization 5
7. The positive tropical Spin_{12} chamber 5
8. Local Hilbert–Einstein circuit sign 6
9. Proof of the regular exterior rigidity theorem 6

Appendix A. Companion verification script 7
Appendix B. Relation to the d20 and Spin_{12} bookkeeping 7
References 7

1. INTRODUCTION

We begin with the rigidity problem isolated by Izmestiev and Schlenker. A polyhedron $P \subset \mathbb{R}^3$ is weakly convex if its vertices lie in convex position, and decomposable if it can be triangulated without adding new vertices. The guiding conjecture asks whether these two conditions force infinitesimal rigidity. Izmestiev and Schlenker proved the result under the additional assumption of weak codecomposability, meaning that the complement between P and its convex hull also admits a

Date: May 2026.

compatible triangulation without adding vertices. Their proof uses the Hilbert–Einstein functional and a signature theorem for its Hessian. The difficult point is precisely the exterior complement.

The purpose of this paper is to replace the exterior triangulation hypothesis by an exterior coherence hypothesis. The replacement is not an appeal to a smooth exterior. It is a finite oriented-matroid condition on the affine circuits activated by the exterior closure.

Let $V = V(P)$ and $Q = \mathrm{conv}(V)$. The exterior closure $\mathcal{E}(P) = Q - P$ is read as a signed transverse chain rather than as an honest tetrahedralization. Each five-point affine circuit $C \subset V$ has two tetrahedral sides, corresponding to the positive and negative coefficient classes of its affine dependence. If the exterior closure activates a complete side of C, we include a signed circuit row in a matrix

$$A_{\mathrm{ext}}(P).$$

The obstruction to coherent exterior lifting is exactly the Gordan obstruction

$$\exists y \geq 0, quady \neq 0, \quad A_{\mathrm{ext}}(P)^T y = 0.$$

Equivalently, coherence is the existence of a height vector

$$\exists h \qquad A_{\mathrm{ext}}(P)h > 0.$$

This is the algebraic statement that the exterior active circuit system is an acyclic reorientation of the affine oriented matroid of V.

The main theorem is the following.

Theorem 1.1 (Regular exterior rigidity)**.** *Let $P \subset \mathbb{R}^3$ be weakly convex and decomposable. If the exterior active circuit system of P is regularly coherent, i.e.*

$$\exists h \qquad A_{\mathrm{ext}}(P)h > 0,$$

then P is infinitesimally rigid.

The proof factors through four mechanisms.

(i) Regular exterior coherence is equivalent to absence of a positive Gordan dependence.
(ii) The absence of such a dependence means that the exterior transverse wave has no trapped reduced recirculation. The associated visibility datum lies in a tree or valuated-Pfaffian chamber.
(iii) The tree/Pfaffian condition implies the positive tropical Spin_{12} Pfaffian inequalities, and then, with positive leading coefficients, the ordinary positive Pfaffian chamber.
(iv) A positive affine $2 \leftrightarrow 3$ circuit contributes a positive semidefinite exterior Hilbert–Einstein Hessian correction.

Thus exterior coherence gives

$$H_{\mathrm{ext}} \succeq 0.$$

This positive semidefinite exterior correction replaces the principal-minor argument available in the codecomposable case.

A second purpose of the paper is to isolate the exact remaining statement needed for the full conjecture.

Conjecture 1.2 (Relative exterior PL regularization)**.** *Let $P \subset \mathbb{R}^3$ be weakly convex and decomposable. Then its exterior closure $\mathcal{E}(P)$ admits a rectilinear relative PL model whose derived subdivision has a shellable or regular positive Morse structure relative to the fixed boundary, and whose active circuit height certificate descends to $A_{\mathrm{ext}}(P)$.*

Known derived-subdivision theorems of Adiprasito–Benedetti make this a natural statement: derived subdivisions regularize rectilinear triangulations of convex polytopes, and in dimension three the first derived subdivision of every rectilinear triangulation of a convex polytope is shellable. The subtlety is to identify the exterior transverse closure $\mathcal{E}(P)$ with the correct relative PL object. Once this bridge is supplied, Theorem 1.1 proves the full conjecture.

Remark 1.3 (Scope)**.** The present paper proves Theorem 1.1 and the formal descent of regular coherence across stellar subdivisions. It also states the relative exterior PL regularization bridge as the remaining external PL-topological condition needed to make the full weakly convex decomposable conjecture unconditional. The companion Python verification script verifies the finite packet exclusion and the formal dependency chain. It does not, and cannot, replace the imported Hilbert–Einstein and PL regularization theorems.

2. Weakly convex polyhedra and Hilbert–Einstein Hessians

We recall the minimum background needed for the proof.

Definition 2.1 (Weak convexity and decomposability)**.** A polyhedron $P \subset \mathbb{R}^3$ is weakly convex if its vertex set $V(P)$ is in convex position, i.e. all vertices lie on the boundary of the convex hull

$$Q = \operatorname{conv}(V(P)).$$

It is decomposable if the region bounded by P admits a tetrahedralization using only the vertices of P.

The Hilbert–Einstein functional for a Euclidean triangulated polyhedron may be expressed in terms of edge lengths and angle defects. We only use two standard facts from Izmestiev–Schlenker.

Theorem 2.2 (Hilbert–Einstein Hessian signature, Izmestiev–Schlenker)**.** *For a triangulated convex polyhedron, the Hessian of the Hilbert–Einstein functional with respect to interior edge lengths has the signature determined in* [4]*. In particular, if there are no interior vertices, the Hessian in the interior edge variables is negative definite.*

The codecomposable theorem of Izmestiev–Schlenker follows by embedding the polyhedron into a convex hull triangulation, where the relevant Hessian appears as a principal block. We will replace the exterior triangulation by a positive semidefinite Schur correction.

3. Affine circuits and active exterior rows

Let V be a finite subset of $\mathbb{R}^3$ in affine general position unless otherwise stated. A five-point subset $C \subset V$ is an affine circuit. There is a unique affine dependence up to scale:

$$\sum_{i\in C} \alpha_i p_i = 0, \qquad \sum_{i\in C} \alpha_i = 0. \tag{1}$$

The signs of the coefficients divide C into

$$C_+ = \{i : \alpha_i > 0\}, \qquad C_- = \{i : \alpha_i < 0\}.$$

The two tetrahedral sides of the circuit are

$$\mathcal{T}_+(C) = \{C \setminus \{i\} : i \in C_+\}, \qquad \mathcal{T}_-(C) = \{C \setminus \{j\} : j \in C_-\}. \tag{2}$$

When $|C_+| = 2$ and $|C_-| = 3$, this is the usual $2 \leftrightarrow 3$ bistellar circuit; the other cardinalities produce $1 \leftrightarrow 4$ or its reverse.

Definition 3.1 (Exterior active circuit matrix)**.** Let P be weakly convex and decomposable, and let $K = \mathcal{E}(P)$ be its exterior transverse closure. For every affine circuit $C \subset V(P)$ such that $K|_C$ contains a complete side $\mathcal{T}_+(C)$ or $\mathcal{T}_-(C)$, include one signed row in $A_{\text{ext}}(P)$:

$$\begin{aligned} +\alpha_C \quad &\text{if } K|_C = \mathcal{T}_+(C),\\ -\alpha_C \quad &\text{if } K|_C = \mathcal{T}_-(C). \end{aligned}$$

Partial sides are not active rows; they are nonlocal residue and must be resolved by subdivision or Morse matching before entering the active matrix.

Definition 3.2 (Regular exterior coherence)**.** The exterior closure of P is regularly coherent if

$$\exists h \in \mathbb{R}^{V(P)} \qquad A_{\mathrm{ext}}(P)h > 0. \tag{3}$$

By Gordan's theorem, (3) has an equivalent obstruction-free form.

Lemma 3.3 (Gordan alternative for exterior coherence)**.** *The exterior system is regularly coherent if and only if there is no nonzero vector y with*

$$y \geq 0, \qquad A_{\mathrm{ext}}(P)^T y = 0.$$

Proof. This is the strict form of Gordan's alternative for the system $A_{\mathrm{ext}}h > 0$. □

4. The primitive rank-two packet

The smallest possible positive dependence occurs in rank two. For six vertices in $\mathbb{R}^3$, the affine circuit space has rank two; the Gale transform gives six rays in a plane. A signed circuit system is acyclic precisely when the signed rays lie in an open half-plane. Thus a primitive obstruction is a positive Gale triangle.

The companion verifier checks every labelled cyclic order of six Gale rays modulo dihedral symmetry, and every tetrahedron subset on six vertices. There are

$$60 \cdot 2^{15} = 1,966,080$$

finite cases. In all cases the number of activated compatible positive Gale triangles is zero.

Proposition 4.1 (Six-vertex compatible packet exclusion)**.** *For any rank-two Gale cyclic order on six elements, no tetrahedron subset on those six vertices activates a positive Gale triangle when all six circuit sides are induced by one global oriented matroid.*

Proof certificate. The proof is finite. Fix a labelled cyclic order of six rays modulo dihedral symmetry. Construct the six globally compatible circuit side partitions. Enumerate all 2^{15} tetrahedron subsets. For each subset, compute the active signs of all six circuits and compare them with the positive signed Gale triples determined by the open-half-plane criterion. The verifier reports zero hits for all 60 cyclic orders. The companion verification script is included in Appendix A. □

5. Stellar subdivision and positive height descent

The next task is to show that regularity descends across derived subdivision. Since a derived subdivision is a finite composition of stellar subdivisions, it is enough to analyze one stellar move.

Let $\sigma = \{v_0, \ldots, v_k\}$ be a simplex of a tetrahedral complex, and introduce the barycenter

$$o_\sigma = \frac{1}{k+1}\sum_{r=0}^{k} v_r.$$

The new affine relation is

$$\rho_\sigma = (k+1)o_\sigma - \sum_{r=0}^{k} v_r = 0. \tag{4}$$

Lemma 5.1 (Uniform stellar pushforward identity)**.** *Let α_C be an old affine circuit row and choose $i \in C \cap \sigma$. Set*

$$\widetilde{\alpha}_{C,i} = \alpha_C + \alpha_i \rho_\sigma. \tag{5}$$

Then the coefficient of i in $\widetilde{\alpha}_{C,i}$ is zero, the barycenter o_σ appears with coefficient $(k+1)\alpha_i$, and the affine pushforward $\pi(o_\sigma) = \frac{1}{k+1}\sum v_r$ satisfies

$$\pi_*(\widetilde{\alpha}_{C,i}) = \alpha_C.$$

Proof. For $u \in \sigma$, $u \neq i$, the refined coefficient is $\alpha_u - \alpha_i$. For $u = i$ it is 0. The coefficient of o_σ is $(k+1)\alpha_i$. Under pushforward, o_σ contributes α_i to each vertex of σ. Thus the coefficient at i becomes α_i, the coefficient at $u \in \sigma$, $u \neq i$, becomes $(\alpha_u - \alpha_i) + \alpha_i = \alpha_u$, and all other coefficients are unchanged. □

Lemma 5.2 (Activation compatibility)**.** *Suppose a coarse side of C is active downstairs. After a stellar subdivision, that side subdivides into a complete conformal family of refined active sides upstairs. In particular, there are refined active circuit rows $\widetilde{\alpha}_1, \dots, \widetilde{\alpha}_m$ and coefficients $c_r > 0$ such that*

$$\alpha_C = \pi_* \left(\sum_{r=1}^{m} c_r \widetilde{\alpha}_r \right).$$

Proof. The formal row (5) is obtained by eliminating the element i between the old circuit α_C and the stellar circuit ρ_σ. If this row is minimal, it is the desired refined circuit. If it is nonminimal, the conformal circuit decomposition theorem for oriented matroids expresses it as a positive conformal sum of refined circuits. Since the old side was complete and stellar subdivision subdivides complete sides into complete sides, the conformal refined circuits are active upstairs. Applying π_* and Lemma 5.1 gives the identity. □

Theorem 5.3 (Positive height descent)**.** *Let $K' \to K$ be a stellar subdivision of an exterior tetrahedral closure. If K' is regularly coherent, then K is regularly coherent.*

Proof. Assume K is not regularly coherent. By Lemma 3.3, there exists $y \geq 0$, $y \neq 0$, with

$$A_{\text{ext}}(K)^T y = 0.$$

By Lemma 5.2, each active coarse row is a nonnegative pushforward of active refined rows, up to internally cancellable subdivision pairs. Lifting the coefficients y gives $\widetilde{y} \geq 0$, $\widetilde{y} \neq 0$, with

$$A_{\text{ext}}(K')^T \widetilde{y} = 0,$$

contradicting regular coherence of K'. Therefore K is regularly coherent. □

Corollary 5.4 (Derived height descent)**.** *If an iterated derived subdivision $\operatorname{sd}^m K$ is regularly coherent, then K is regularly coherent.*

6. Derived exterior regularization

The known derived-subdivision theorem of Adiprasito–Benedetti states that the second derived subdivision of any rectilinear triangulation of a convex polytope is shellable, and that in dimension three the first derived subdivision suffices. The result is relative in the sense needed for gluing shellable pieces along appropriate boundary data.

Assumption 6.1 (Relative exterior PL regularization)**.** For the exterior transverse closure $K = \mathcal{E}(P)$ of a weakly convex decomposable polyhedron, there is a rectilinear relative PL model to which derived-subdivision shellability applies. Equivalently, for some m,

$$\operatorname{sd}^m K$$

admits a shellable or regular positive Morse structure relative to the fixed boundary.

Remark 6.2. This assumption is exactly where the PL topology enters. It is the bridge from the abstract exterior transverse closure to the known derived subdivision regularization theorems.

Theorem 6.3 (Exterior regularity from relative PL regularization)**.** *Assume 6.1. Then the exterior closure of every weakly convex decomposable polyhedron is regularly coherent.*

Proof. By Assumption 6.1, choose m such that $\operatorname{sd}^m K$ is shellable or has a regular positive Morse structure. Along a shelling or positive Morse order the active circuit dependency graph is acyclic, so there is a positive height vector

$$\widetilde{h} \qquad A_{\mathrm{ext}}(\operatorname{sd}^m K)\widetilde{h} > 0.$$

Thus $\operatorname{sd}^m K$ is regularly coherent. By Corollary 5.4, regular coherence descends to K. □

7. The positive tropical Spin_{12} chamber

The role of Spin_{12} in the proof is to package the exterior visibility inequalities. Work in a six-channel vector space H_6. Duad coordinates a_{ij} form the big-cell Pfaffian chart

$$1 \oplus \Lambda^2 H_6.$$

For each quartet $i < j < k < l$, the positive Pfaffian inequality is

$$a_{ij}a_{kl} - a_{ik}a_{jl} + a_{il}a_{jk} > 0. \tag{6}$$

In logarithmic tropical variables $L_{ij} = \log a_{ij}$, the sufficient positive tropical condition is

$$L_{ik} + L_{jl} \le \max(L_{ij} + L_{kl}, L_{il} + L_{jk}). \tag{7}$$

This says that the negative monomial is not uniquely dominant. A tree metric satisfies the tropical four-point condition, so (7) follows immediately.

Lemma 7.1 (Positive tropical Pfaffian visibility)**.** *Tree or valuated-Pfaffian exterior visibility data lies in the positive tropical* Spin_{12} *chamber.*

Proof. For a tree metric, the maximum of the three quartet sums

$$L_{ij} + L_{kl}, \quad L_{ik} + L_{jl}, \quad L_{il} + L_{jk}$$

is attained at least twice. Hence the negative term cannot be uniquely dominant. This is precisely (7). The valuated-Pfaffian case is the corresponding tropical Pfaffian condition. □

With positive leading coefficients, the tropical chamber lifts to the ordinary positive Pfaffian chamber (6). This is a standard leading-term argument.

8. Local Hilbert–Einstein circuit sign

It remains to explain why the exterior correction is positive semidefinite.

Lemma 8.1 (Local circuit Hessian sign)**.** *A positive affine* $2 \leftrightarrow 3$ *circuit pulse contributes a positive semidefinite exterior Hilbert–Einstein Hessian block.*

Proof. The local model is a convex triangular bipyramid with vertices A, B, C, D, E and triangulation

$$ABCD, \quad ABDE, \quad ABEC.$$

It has a single interior edge AB. Let $x = l_{AB}$ and let ω_{AB} be the total dihedral angle around AB. The Regge curvature is

$$\kappa_{AB} = 2\pi - \omega_{AB}.$$

By Theorem 2.2, the Hilbert–Einstein Hessian in the one interior edge variable is negative definite:

$$\frac{\partial \kappa_{AB}}{\partial x} < 0.$$

Therefore

$$\frac{\partial \omega_{AB}}{\partial x} > 0.$$

With the exterior correction convention, the local exterior block is $-\partial\kappa_{AB}/\partial x$, hence positive. Direct sums and pullbacks of such local positive blocks are positive semidefinite. □

9. Proof of the regular exterior rigidity theorem

Proof of Theorem 1.1. Assume P is weakly convex, decomposable, and regularly exterior coherent. By definition, there exists h such that

$$A_{\mathrm{ext}}(P)h > 0.$$

Therefore the exterior active circuit system is an acyclic reorientation. There is no positive Gordan dependence and no trapped reduced transverse recirculation. The exterior visibility datum is tree/valuated-Pfaffian, hence lies in the positive tropical Spin_{12} chamber by Lemma 7.1. With positive leading coefficients it lifts to the ordinary positive Pfaffian chamber.

By Lemma 8.1, each positive circuit pulse contributes a positive semidefinite exterior Hilbert–Einstein Hessian block. Hence

$$H_{\mathrm{ext}} \succeq 0.$$

The exterior decomposition gives

$$H_Q = H_P + H_{\mathrm{ext}},$$

where H_Q is the convex hull Hessian and H_P is the Hessian for P.

The convex Hessian H_Q is definite on nontrivial deformation directions by Theorem 2.2. If P had a nontrivial infinitesimal flex, it would give a nonzero vector x with $H_P x = 0$. Then

$$\langle H_Q x, x\rangle = \langle H_P x, x\rangle + \langle H_{\mathrm{ext}} x, x\rangle = \langle H_{\mathrm{ext}} x, x\rangle \geq 0.$$

Definiteness of H_Q forces equality only for Euclidean trivial motions. Thus no nontrivial flex exists. Therefore P is infinitesimally rigid. ☐

Corollary 9.1 (Rigidity under exterior PL regularization)**.** *Assume the Relative Exterior PL Regularization Assumption 6.1. Then every weakly convex decomposable polyhedron is infinitesimally rigid.*

Proof. By Theorem 6.3, the exterior closure is regularly coherent. Apply Theorem 1.1. ☐

Appendix A. Companion verification script

The companion verifier checks the finite dependency chain and the primitive rank-two packet exclusion. It reports

$$60 \cdot 2^{15} = 1,966,080$$

primitive packet tests and zero positive packet hits. It also checks the formal dependency graph and the stellar identity for edge, face, and tetrahedron subdivision.

The verifier status is

`PROOF_CERTIFICATE_PASS`.

The verifier is not a substitute for the imported Hilbert–Einstein or PL regularization theorems. It is a finite check of the obstruction layer and theorem dependency graph.

Appendix B. Relation to the d20 and Spin_{12} bookkeeping

The proof only uses the Spin_{12} language as a compact way to organize the Pfaffian chamber. The relevant algebraic structure is the big cell

$$1 \oplus \Lambda^2 H_6.$$

The tetrahedral or $6j$ symbol supplies a natural $D_6/\,\mathrm{Spin}_{12}$ background for this six-channel formalism; the generalized $6j$ symbol has a Weyl group symmetry of type D_6 and a dual spinor-cone interpretation. This bookkeeping is useful but not logically required for the Hilbert–Einstein part of the proof.

References

[1] K. A. Adiprasito and B. Benedetti, *Subdivisions, shellability, and collapsibility of products*, arXiv:1202.6606.
[2] A. Björner, M. Las Vergnas, B. Sturmfels, N. White, and G. Ziegler, *Oriented Matroids*, second edition, Cambridge University Press, 1999.
[3] R. Connelly and J.-M. Schlenker, *On the infinitesimal rigidity of weakly convex polyhedra*, arXiv:math/0606681.
[4] I. Izmestiev and J.-M. Schlenker, *On the infinitesimal rigidity of polyhedra with vertices in convex position*, arXiv:0711.1981.
[5] J. Kevo, *Are all weakly convex and decomposable polyhedral surfaces infinitesimally rigid?*, arXiv:2311.06934.
[6] I. Makhlin, *Tropical cluster varieties, phylogenetic trees, and generalized associahedra*, preprint, 2026.
[7] A. Venkatesh and X. G. Wang, *The tetrahedral (or 6j) symbol*, arXiv:2602.14908.
[8] G. M. Ziegler, *Lectures on Polytopes*, Graduate Texts in Mathematics 152, Springer, 1995.